LEI CHENG

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RESEARCH INTERESTS

My research focuses on autonomous systems and Physical AI that operate in the real world by integrating multimodal sensing and sensor fusion, deep perception, AI-based scene understanding and reasoning, and intelligent decision-making and action. My goal is to develop robust, adaptive, and trustworthy autonomous systems capable of understanding complex real-world environments and acting intelligently within them.

Research Areas: Physical AI · Autonomous Systems · Multimodal AI · Robotics · Sensor Fusion · Deep Perception

CURRENT POSITION

Assistant Professor **Aug 2026 – Present**
Department of Electrical and Computer Engineering
The University of Texas Rio Grande Valley, Edinburg, USA
Director, Perception, AI, and Autonomy (PAIA) Lab

PREVIOUS APPOINTMENTS

Faculty Assistant **Aug 2025 – Aug 2026**
Department of Civil and Environmental Engineering(CEE)–Maryland Transportation Institute(MTI)
University of Maryland, College Park, USA
Supervisor: Prof. Xianfeng Yang

Research Member **Jul 2019 – May 2021**
Artificial Intelligence Research Center
Peng Cheng Laboratory, Shenzhen, China

EDUCATION

University of Arizona, Tucson, USA **Aug 2021 – Aug 2025**
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Siyang Cao

Peking University, Beijing, China **Aug 2016 – Jul 2019**
M.Eng. in Integrated Circuit Engineering
Advisor: Prof. Wei Yan

Northeast Petroleum University, Daqing, China **Aug 2012 – Jul 2016**
B.S. in Applied Physics

RESEARCH PROJECT EXPERIENCE

BEACON: Wearable Multi-Sensor Safety System for Transportation in High-Risk Environments **Sep 2026 – Present**
USDOT, University Transportation Centers Program – University Transportation Center for Railway Safety
- Role: Co-Principal Investigator (**Co-PI**). PI: Jazmin Ley; Co-PIs: Weiguang Yang, and Constantine Tarawneh.
- Funding: **$150,000**.

Invasive Species Control **Dec 2025 – Present**
Maryland Department of Transportation State Highway Administration FY 2026 Research Work Program
- Role: Principal Investigator (**PI**). Co-PIs: Yaobang Gong and Daniel Buonaiuto.
- Funding: **$152,241**.

Real-time Work Zone Speed Management with Multi-source Data **Aug 2025 – Aug 2026**
SMART GRANT PROGRAM of USDOT and the Maryland-DOT State Highway Administration

- Developed an unmanned aerial vehicles (UAV)-based deep learning traffic monitoring system to enhance work zone safety and operational efficiency.
- Implemented a UAV-Enhanced Variable Speed Limit (VSL) system using adaptive advisory speed signs to smooth approaching traffic and mitigate bottlenecks, complemented by a UAV-Powered Advanced Safety Warning (ASW) system that alerts workers to traffic anomalies in real time.

Advanced Artificial Intelligence Research for Equitable Safety of Vulnerable Road Users **Jan 2024 – Aug 2025**
The Federal Highway Administration Exploratory Advanced Research Program *Teamed with UTC and USFCA*

- Investigated LiDAR object detection algorithms and the sensor fusion techniques involving LiDAR.
- Developed an online automatic calibration method for LiDAR-camera systems, leveraging PointNet++ and ResNet to match common objects between LiDAR and camera data. The method further incorporates two post-refinement techniques: iterative optimization and a cross-attention-based deep learning model, to enhance calibration performance.

Deep Learning-Enhanced Radar Object Detection and Sensor Fused Object Tracking **Aug 2023 – Jan 2024**
Graduate Assistantship from National Institutes of Health (NIH)

- Developed TransRAD, a 3D radar detection model using Retentive Vision Transformer and Retentive Manhattan Self-Attention to encode spatial priors for accurate RAD-space detection.
- Proposed a radar-camera multi-object tracking framework that minimizes manual intervention by leveraging online targetless calibration and common features to integrate radar and camera detections seamlessly.

Deep Learning-Based Detection of Personal Protective Equipment Compliance in Mining **Aug 2023 – Dec 2023**
Funded by Mine Automation and Autonomous Systems (MAAS) Laboratory *Prof. Nathalie Risso*

- Contributed to the collection and processing of a real-world PPE dataset gathered at the San Xavier Mining Site.
- Developed a PPE detection system integrating YOLOv8 and a self-attention mechanism, effectively identifying compliance with helmet, vest, mask, and goggles usage.

Radar-Camera Online Calibration Based on the Common Features of Radar and Camera **Apr 2023 – Jul 2023**
Funded by National Institute for Transportation and Communities, and Tucson Department of Transportation

- Developed a novel deep learning model to learn and extract common features from raw radar data and camera images.
- Developed a feature-based online targetless calibration method that leverages the extracted common feature to enable real-time calibration.

Deep Neural Network-based Intelligent RF Sensing for Target Detection and Tracking **Aug 2021 – Apr 2023**
Sony Research Award Program

- Proposed a flexible, easy-to-use and accurate method for extrinsic calibration of 3D radar and camera.
- Developed and implemented a real-time object tracking system based on radar-camera fusion using ROS.
- Developed RADYolo, a deep learning-based radar object detector that improves object detection accuracy.
- Created a deep learning-based multi-object tracking framework that combines data from radar and camera sensors for more accurate and reliable tracking, with GNSS data as the ground-truth.

PUBLICATIONS

Book Chapters

[1] **Lei Cheng**, Siyang Cao, Shunqiao Sun, and Vishal Patel. "Deep Learning Based Automotive Radar Sensor Fusion." Book: Radar Machine Learning for Autonomous Driving, Artech House, 2025. *(Under Publication)*

Journal Papers

[1] **Lei Cheng**, and Siyang Cao. "Heterogeneous Sensor Fusion for Environmental Perception in Autonomous Systems: Principles, Frameworks, and Emerging Trends." IEEE Transactions on Intelligent Transportation Systems, 2026. *(Under Review)*

[2] **Lei Cheng**, Lihao Guo, Tianya Zhang, Tam Bang, Austin Harris, Mustafa Hajij, Mina Sartipi, and Siyang Cao. "CalibRefine: Deep Learning-Based Online Automatic Targetless LiDAR–Camera Calibration with Iterative and Attention-Driven Post-Refinement." IEEE Transactions on Instrumentation and Measurement, 2025.

[3] **Lei Cheng**, and Siyang Cao. "Radar-Camera Fused Multi-Object Tracking: Online Calibration and Common Feature." IEEE Transactions on Intelligent Transportation Systems, 2025.

[4] **Lei Cheng**, and Siyang Cao. "TransRAD: Retentive Vision Transformer for Enhanced Radar Object Detection." IEEE Transactions on Radar Systems, 2025.

[5] Tianya Zhang, **Lei Cheng**, Tam Bang, Lihao Guo, Mustafa Hajij, Siyang Cao, Austin Harris, and Mina Sartipi. "Roadside Sensor Systems for Vulnerable Road User Protection: A Review of Methods and Applications." IEEE Access, 2025.

[6] **Lei Cheng**, Arindam Sengupta, and Siyang Cao. "Deep Learning-Based Robust Multi-Object Tracking via Fusion of mmWave Radar and Camera Sensors." IEEE Transactions on Intelligent Transportation Systems, 2024.

[7] Arindam Sengupta, **Lei Cheng** *(*co-first author)*, and Siyang Cao. "Robust multiobject tracking using mmwave radar-camera sensor fusion." IEEE Sensors Letters, 2022.

Conference Papers

[1] Zihan Ma, Sung Yoon Park, Gang-Len Chang, and **Lei Cheng**. "Design and Field Demonstration of a Real-Time UAS-Based System for Speed Management and Advanced Work Zone Safety Alerts." Transportation Research Board (TRB) Annual Meeting, 2027.

[2] Tam Bang, Hoang H. Nguyen, **Lei Cheng**, Lihao Guo, Siyang Cao, Hussam Abubakr, Tianya Zhang, Austin Harris, and Mina Sartipi. "CLIFE: Camera-LiDAR Fusion Framework for Edge-Deployable Roadside VRU Perception." IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), 2026.

[3] Dianwei Chen, Zifan Zhang, **Lei Cheng**, Yuchen Liu, Xianfeng Yang. "INSIGHT: Enhancing Autonomous Driving Safety through Vision-Language Models on Context-Aware Hazard Detection and Edge Case Evaluation." IEEE Intelligent Vehicles Symposium (IV), 2026.

[4] Tam Bang, Hoang H. Nguyen, Deponker Sarker Depto, **Lei Cheng**, Lihao Guo, Siyang Cao, Tianya Zhang, Austin Harris, and Mina Sartipi. "Real-Time Edge-Based Multi-Sensor Fusion for Vulnerable Road User Safety at Urban Intersections." Transportation Research Board (TRB) Annual Meeting, 2025.

[5] **Lei Cheng**, Arindam Sengupta, and Siyang Cao. "3D Radar and Camera Co-Calibration: A flexible and Accurate Method for Target-based Extrinsic Calibration." IEEE Radar Conference (RadarConf23), 2023.

[6] **Lei Cheng**, and Siyang Cao. "Online Targetless Radar-Camera Extrinsic Calibration Based on the Common Features of Radar and Camera." IEEE National Aerospace and Electronics Conference (NAECON), 2023.

[7] **Lei Cheng**, Wei Yan, Guoqing Xiang, Xiao Liu, and Yunyao Yan. "Fast Sub-pixel Prediction Based on Error Surface Fitting for HEVC." IEEE 4th International Conference on Signal and Image Processing (ICSIP), 2019.

[8] **Lei Cheng**, Wei Yan, Guoqing Xiang, Xiao Liu, and Yunyao Yan. "HEVC fast fractional-pixel prediction with parallel search." In Eleventh International Conference on Digital Image Processing (ICDIP), SPIE, 2019.

[9] Xiao Liu, Wei Yan, Guoqing Xiang, **Lei Cheng** and Yunyao Yan. "A novel fast mode decision algorithm for AVS2 intra coding." In IEEE 4th International Conference on Signal and Image Processing (ICSIP), 2019.

[10] Jianbin Zhao, Yunyao Yan, Xiao Liu, **Lei Cheng**, and Wei Yan. "The Implementation of a Fast RMD Method for Audio Video Coding Standard II (AVS2)." In International Conference on Computer Science and Software Engineering (CSSE), 2018.

TEACHING

EECE 6399: Artificial Intelligence for Autonomous Systems **Fall 2026**

Graduate Course, Department of Electrical and Computer Engineering, UTRGV

- Graduate course on AI methods for autonomous systems: search, constraint satisfaction, MDPs, reinforcement learning, Bayesian networks, machine learning, and language models, with applications to robotics, autonomous driving, and intelligent transportation.

EECE 2106: Digital Systems Engineering I Lab **Fall 2026**

Laboratory Course, Department of Electrical and Computer Engineering, UTRGV

- Laboratory course accompanying Digital Systems Engineering I: design, build, test, and troubleshoot combinational and sequential circuits on breadboards with TTL ICs.

OPEN SOURCE

[1] **CalibRefine**: Deep Learning-Based Online Automatic Targetless LiDAR–Camera Calibration [IEEE TIM, 2025].
GitHub: github.com/radar-lab/Lidar_Camera_Automatic_Calibration

[2] **TransRAD**: Retentive Vision Transformer for Enhanced Radar Object Detection [IEEE TRS, 2025].
GitHub: github.com/radar-lab/TransRAD

[3] **Radar_Camera_MOT**: Radar-Camera Fused Multi-Object Tracking with Online Calibration and Common Feature [IEEE TITS, 2025].
GitHub: github.com/radar-lab/Radar_Camera_MOT

[4] **Online_RC_Calib**: Online Targetless Radar-Camera Extrinsic Calibration Based on Common Features [IEEE NAECON, 2023].
GitHub: github.com/radar-lab/Online-Targetless-Radar-Camera-Extrinsic-Calibration

[5] **3D Radar-Camera Co-Calibration**: Flexible Target-based Extrinsic Calibration for 3D Radar and Camera [IEEE RadarConf, 2023].
GitHub: github.com/radar-lab/sony_project_2022

DISSERTATION

Lei Cheng, *Perception with Sensor Fusion: From Calibration to Object Detection and Tracking* **2025**
Ph.D. Dissertation, The University of Arizona. Advisor: Prof. Siyang Cao.
ProQuest: www.proquest.com/docview/3236320797

PROFESSIONAL ACTIVITIES

- IEEE Transactions on Intelligent Transportation Systems (TITS) **Reviewer**
- IEEE Transactions on Aerospace and Electronic Systems (TAES) **Reviewer**
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) **Reviewer**
- IEEE Transactions on Mobile Computing (TMC) **Reviewer**
- IEEE Transactions on Instrumentation & Measurement (TIM) **Reviewer**
- IEEE Transactions on Radar Systems (TRS) **Reviewer**
- IEEE Robotics and Automation Letters (RA-L) **Reviewer**
- IEEE Internet of Things Journal (IoT) **Reviewer**
- IEEE Sensors Journal **Reviewer**
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) **Reviewer**
- IEEE Signal Processing Letters (SPL) **Reviewer**
- IEEE Sensors Letters **Reviewer**
- IEEE Access **Reviewer**
- IEEE International Conference on Intelligent Transportation Systems (IEE ITSC) **Reviewer**
- IEEE Intelligent Vehicles Symposium (IEEE IV) **Reviewer**
- IEEE Vehicular Technology Conference (IEEE VTC) **Reviewer**
- Information Fusion (INFFUS) **Reviewer**
- Automation in Construction (AUTCON) **Reviewer**
- Elsevier Expert Systems With Applications (ESWA) **Reviewer**
- Elsevier Engineering Applications of Artificial Intelligence (EAAI) **Reviewer**
- Elsevier Robotics and Autonomous Systems (ROBOT) **Reviewer**
- Elsevier Measurement (MEAS) **Reviewer**
- Elsevier Data and Knowledge Engineering (DATAK) **Reviewer**
- Elsevier SoftwareX (SOFTX) **Reviewer**
- Elsevier Smart Agricultural Technology (ATECH) **Reviewer**
- Springer Nature SN Computer Science (SNCS) **Reviewer**
- Institute of Electrical and Electronics Engineers (IEEE) **Graduate Student Member**